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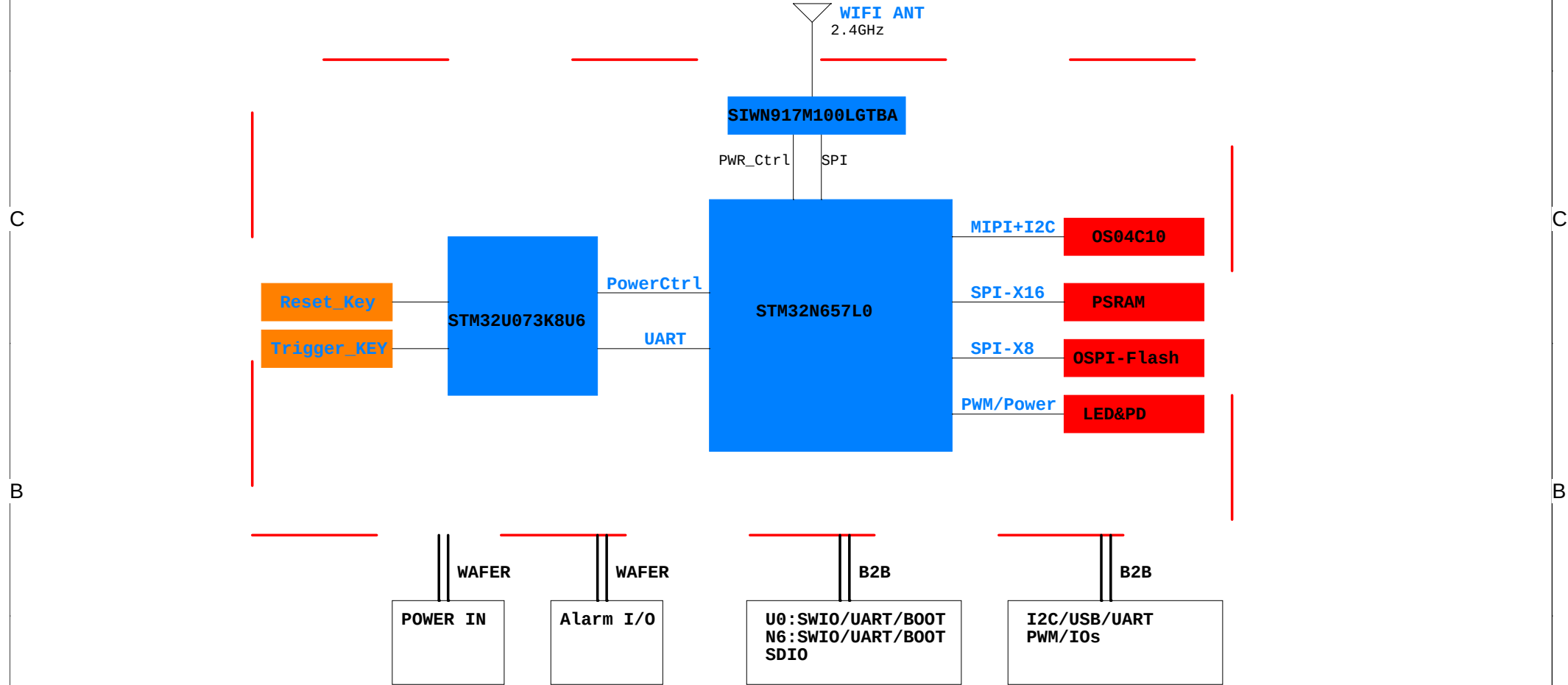
DESIGNED	Tino
REVISER	
REV	V1_1

CHANGE LIST

Revision History

REV. CODE	DATE	Description
V1_0	2026/03/25	New Design
		Ref: U7 change to K2 C119: 0402 change to 0201 0.1uF/6.3V C114: Del USB: Change to USB1
	2026-06-26	1. Add soft-startup res: R88/R89/R90/R92 10K 0402 5% 2. Add R91 100K for PWM_LED (Pull down) 3. OS04C10 CLK source change to Y6: remove R76 0R, add R86 4. Optical Diode:Set to 200Lux Max R52-->12.7K 1%, R55-->15K 1% 5. J3 Pin7 change to PB0 from Boot0

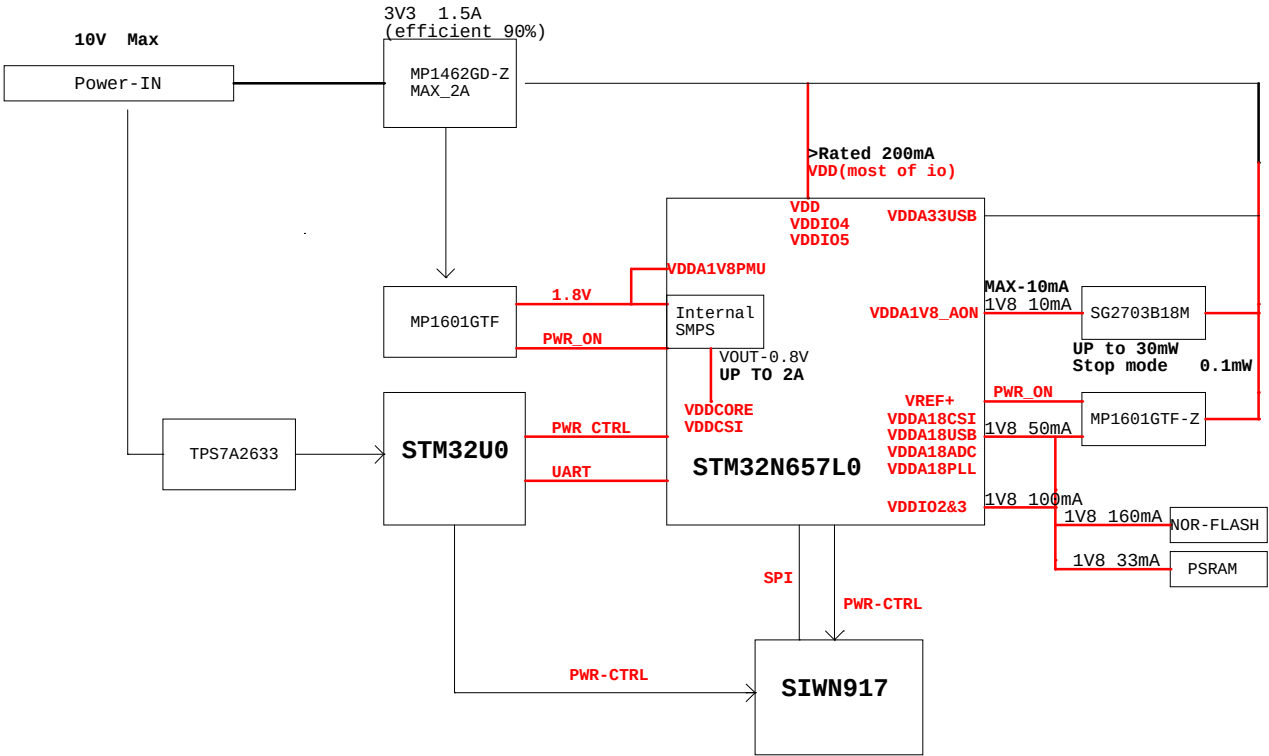
BLOCK DIAGRAM

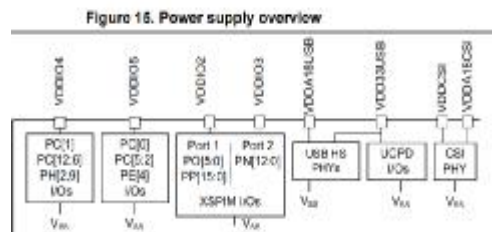
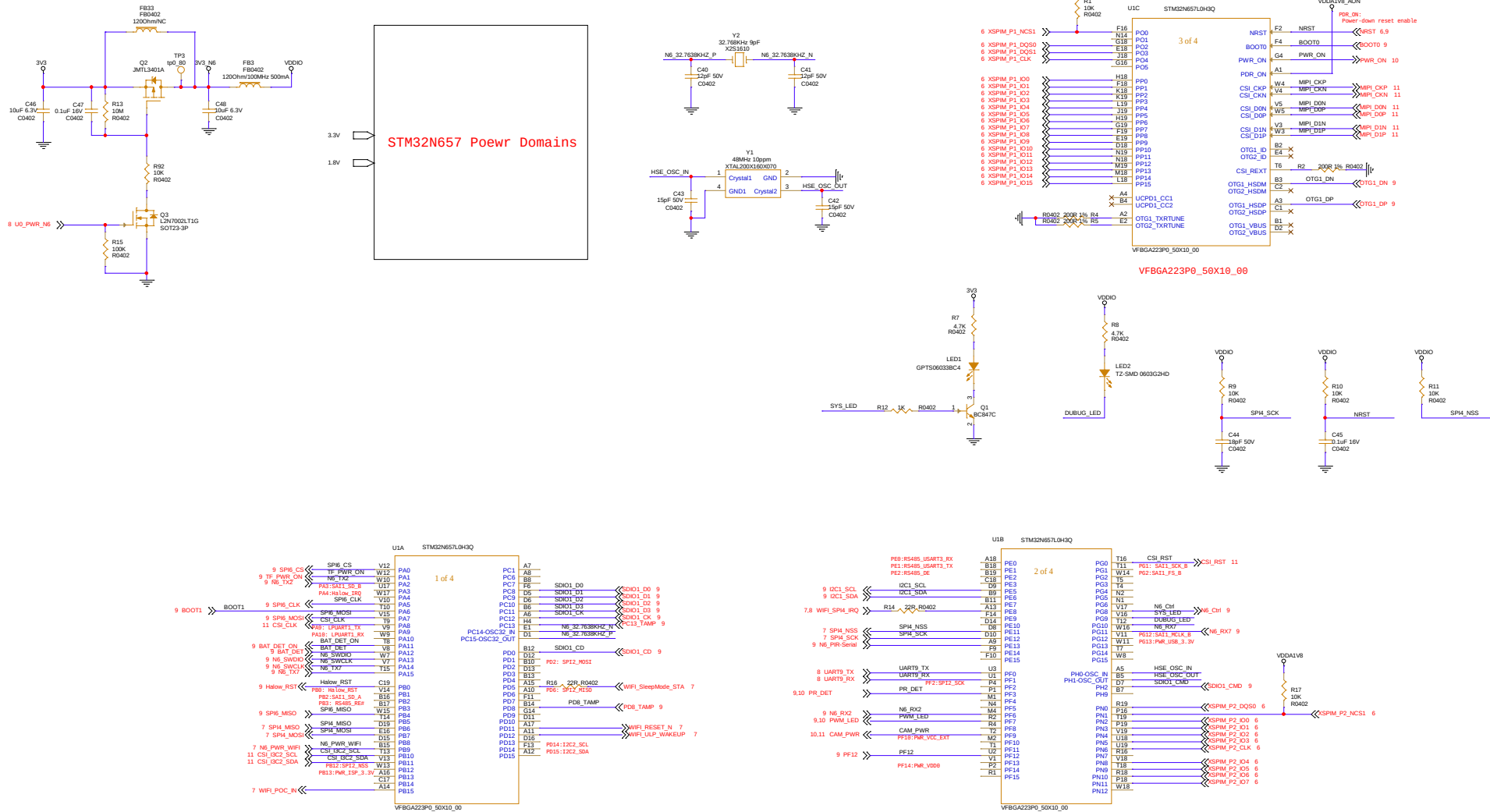


SPI/USB/Power
Cat-1 Module

		https://www.camthink.ai	
Title		BLOCK DIAGRAM	
Size	Project Name	Rev	
A3	ACM12-03-V1.1	V1.1	
Date: Monday, June 29, 2026		Sheet	3 of 11

POWER TREE



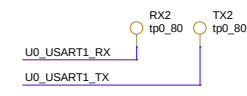
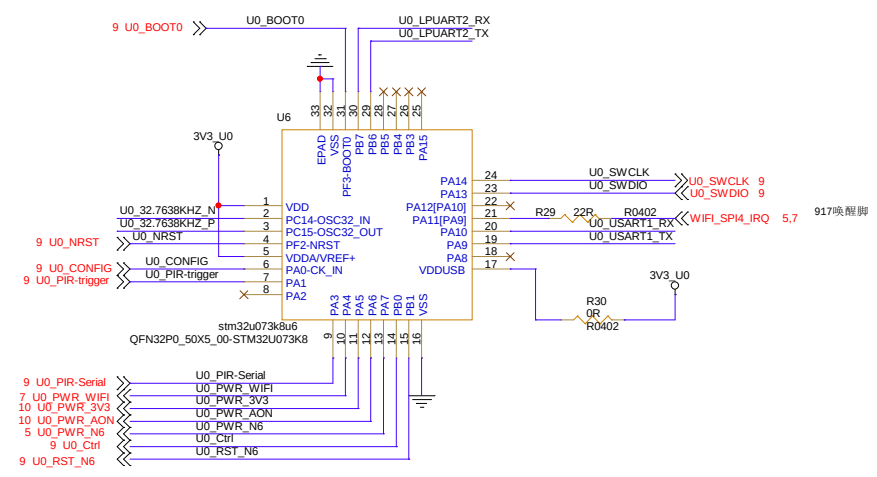
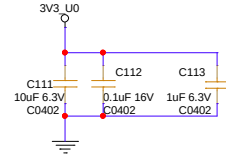
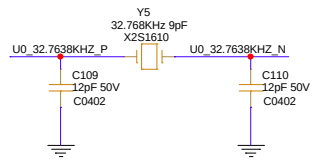
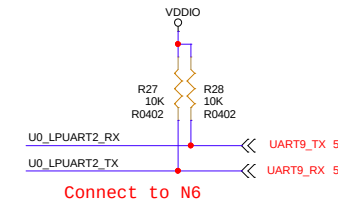
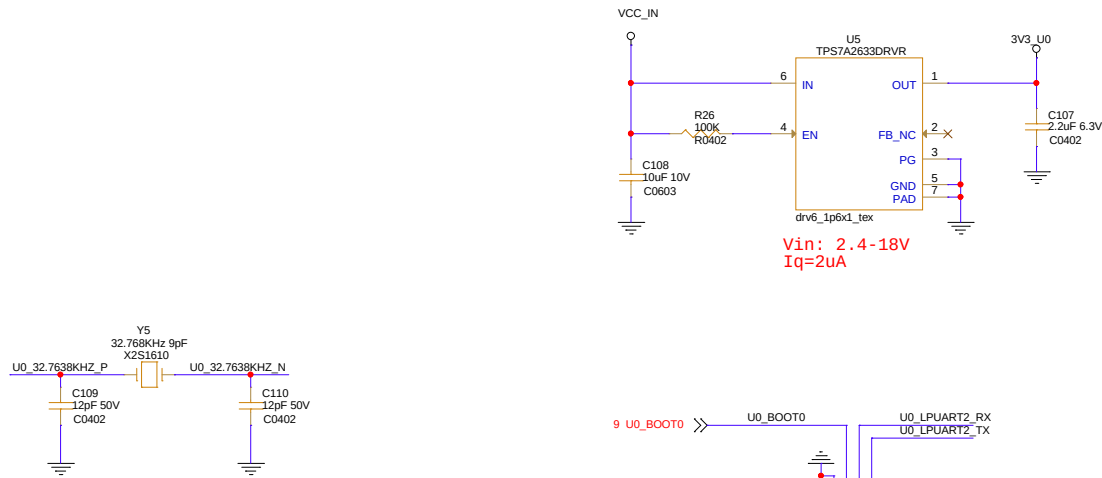


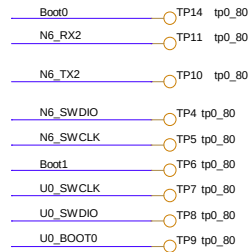
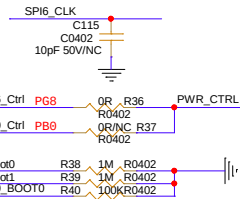
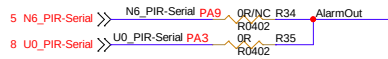
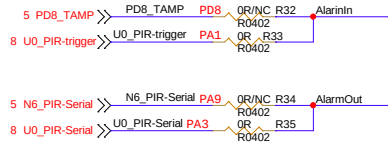
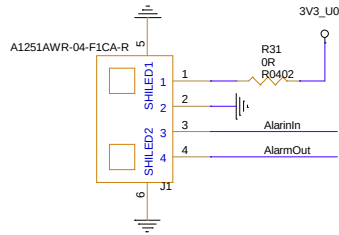
[illegible]

Flash:
1.8V@32mA Max
1.8V@13mA Avg
1.8V@100mA Theory

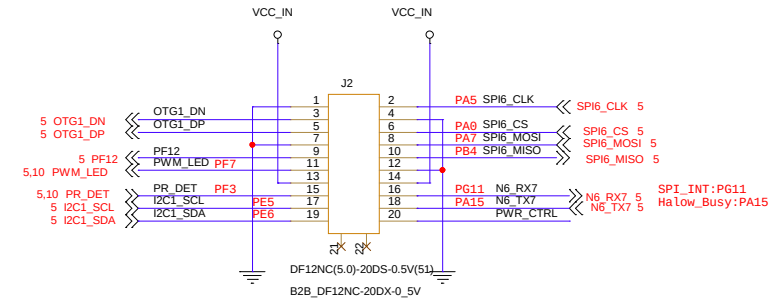
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```
Alt for MX66UM1G45G:
1Gbit: DS-00683-GD55LX01GE
512Mbit: DS-00596-GD25LX512ME  --Default
256Mbit: DS-00514-GD25LX256E
```

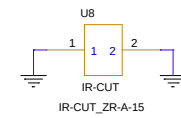
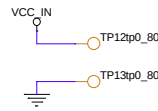
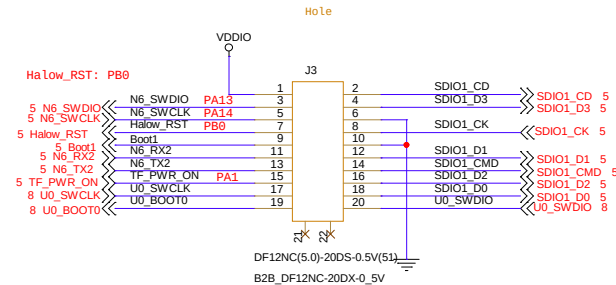





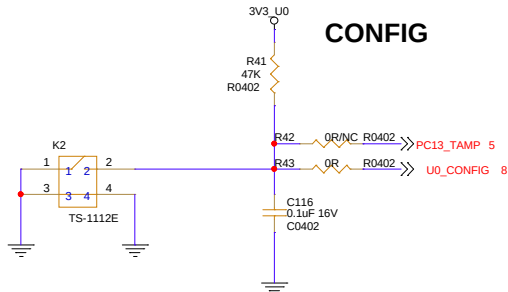
N6 Boot[0:1]
00: Flash
x1: Development
10: Serial Boot (USB/UART)



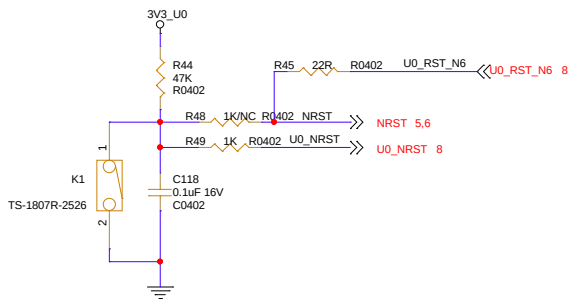
SPI1/2/3/6 Support I2S



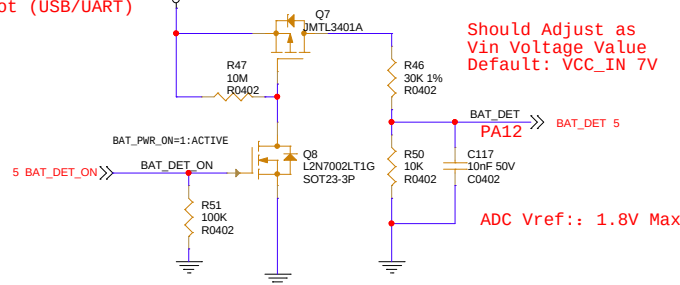
CONFIG



SYS Reset

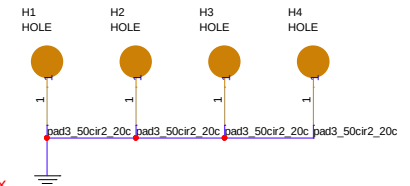


Battery Detection

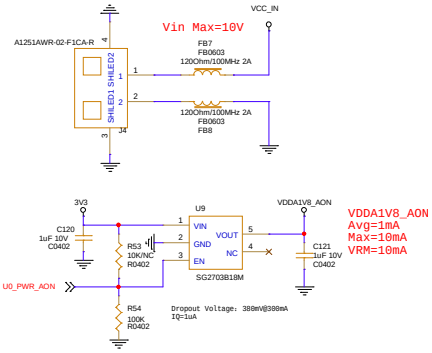


Should Adjust as
Vin Voltage Value
Default: VCC_IN 7V

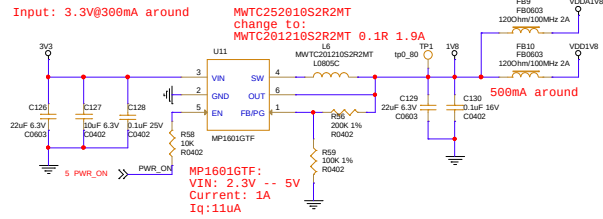
ADC Vref:: 1.8V Max



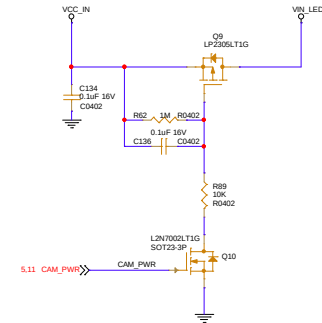
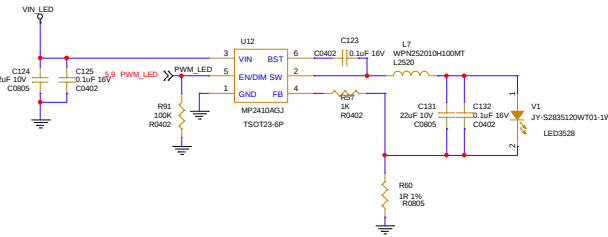
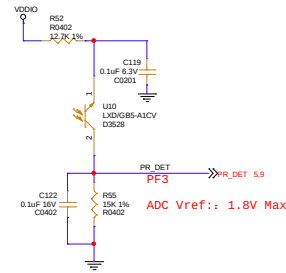
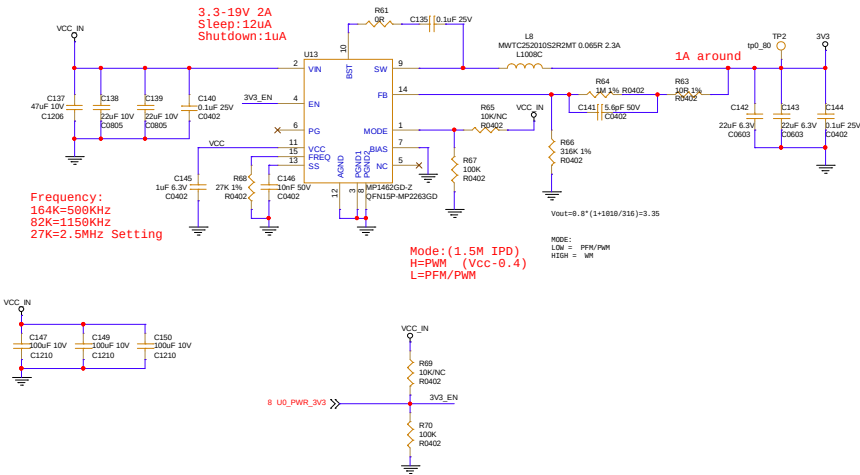
CamThink		https://www.camthink.ai	
Size		Title	
A3		MISC	
Project Name		Rev	
ACM12-03-V1.1		V1.1	
Date: Thursday, July 02, 2026		Sheet 9 of 11	



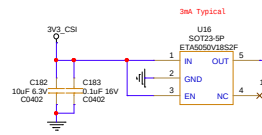
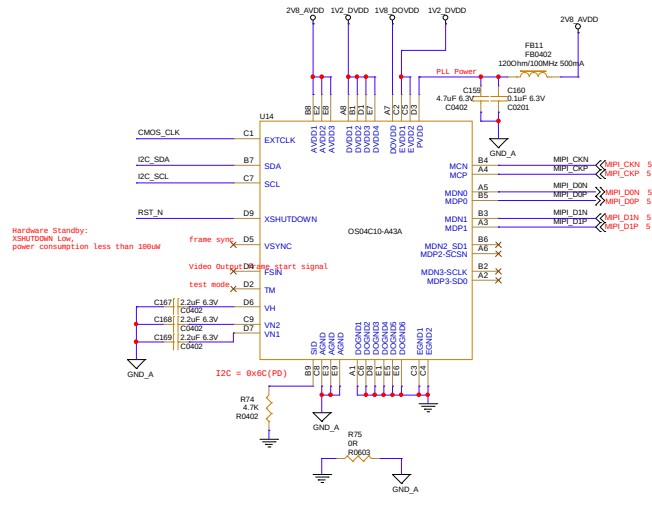
3V3-->1V8



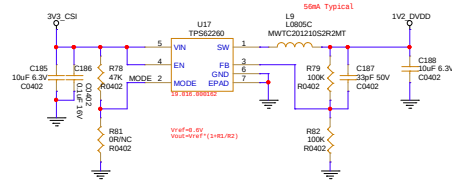
VIN-->3V3



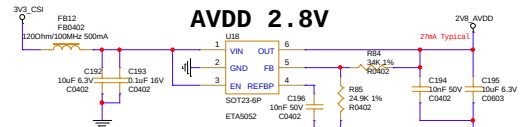
Sensor



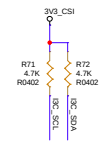
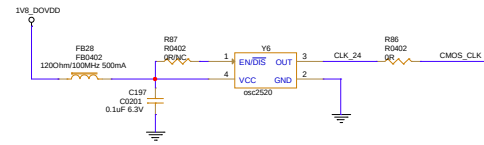
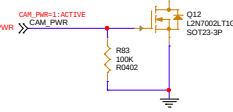
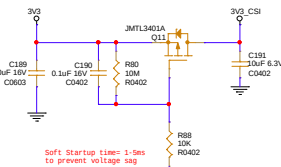
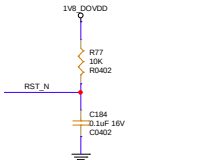
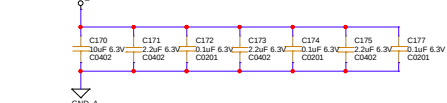
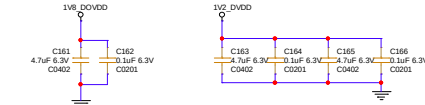
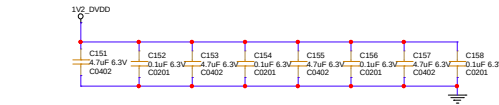
Ven=1.5V



Vout=8.8" (1+1000/800)=1.8
IQ=130uA



Ven=1.2V
Vout=1.2 * (1+R2/R3) = 2.83V



Level conversion

